

Geometric Background

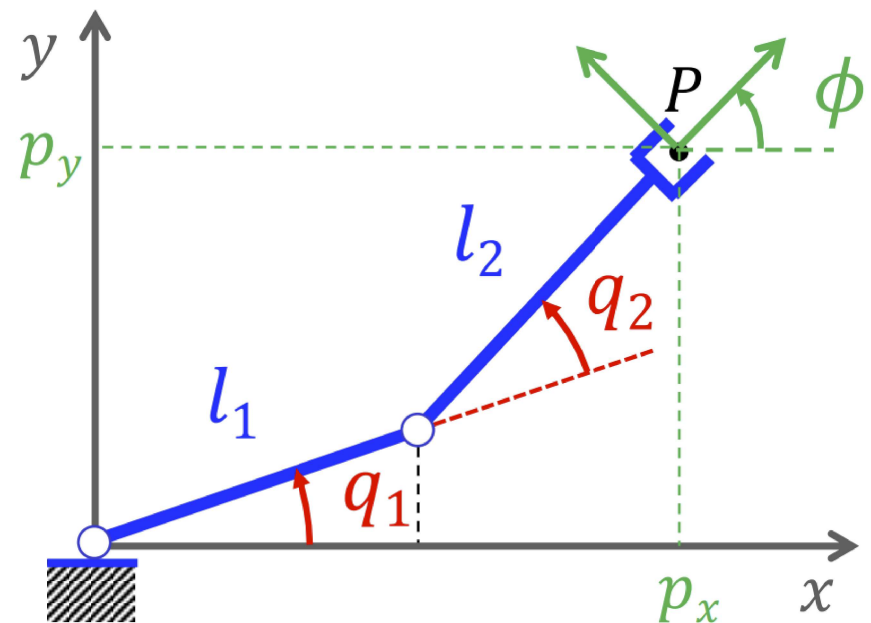
Fundamentals of Robotics

University of Trento

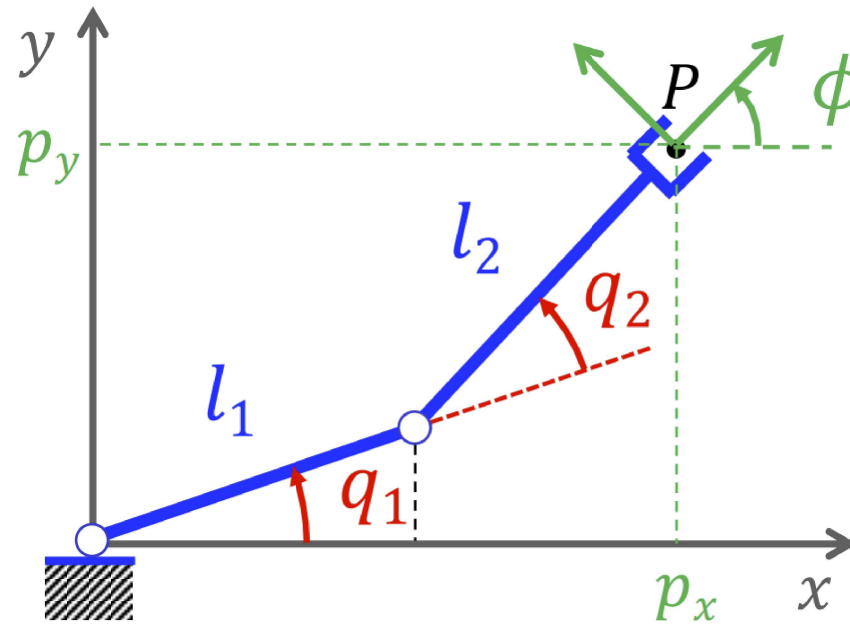
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A simple Planar Robot

- ▶ Consider the simple planar robot in the figure
- ▶ We have got two revolute joints
- ▶ Let q_1 and q_2 be the position of the angles at the two joints
- ▶ We want to find the pose of the end effector $\begin{bmatrix} p_x \\ p_y \\ \phi \end{bmatrix}$ as a function of the joint configuration $\begin{bmatrix} q_1 \\ q_2 \end{bmatrix}$.



A simple Planar Robot (1)



With a little math, we come up with

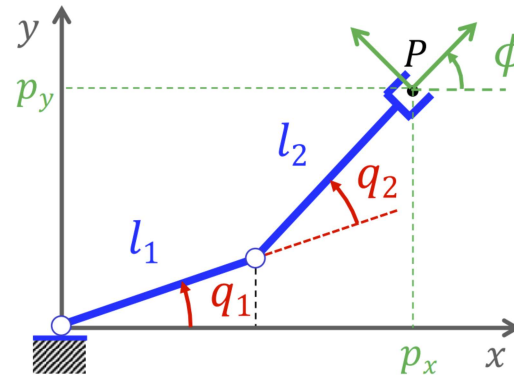
Forward (or direct) kinematics

$$p_x = l_1 \cos(q_1) + l_2 \cos(q_1 + q_2)$$

$$p_y = l_1 \sin(q_1) + l_2 \sin(q_1 + q_2)$$

$$\phi = q_1 + q_2$$

A simple Planar Robot (2)



For inverse kinematics we need a little more math. Since $q_1 + q_2 = \phi$.

$$p_x - l_2 \cos(\phi) = l_1 \cos(q_1)$$

$$p_y - l_2 \sin(\phi) = l_1 \sin(q_1)$$

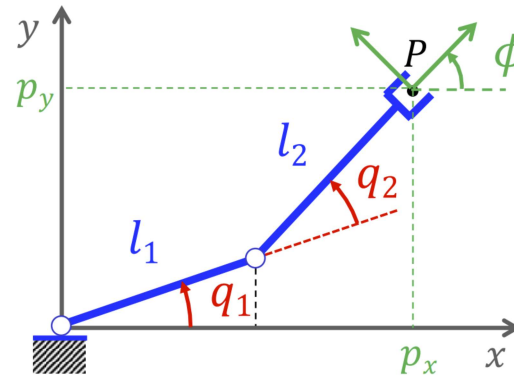
which leads to

Inverse Kinematics

$$q_1 = \arctan_2 \left(\frac{p_x - l_2 \cos(\phi)}{l_1}, \frac{p_y - l_2 \sin(\phi)}{l_1} \right) \quad (1)$$

$$q_2 = \phi - q_1$$

A simple Planar Robot (3)



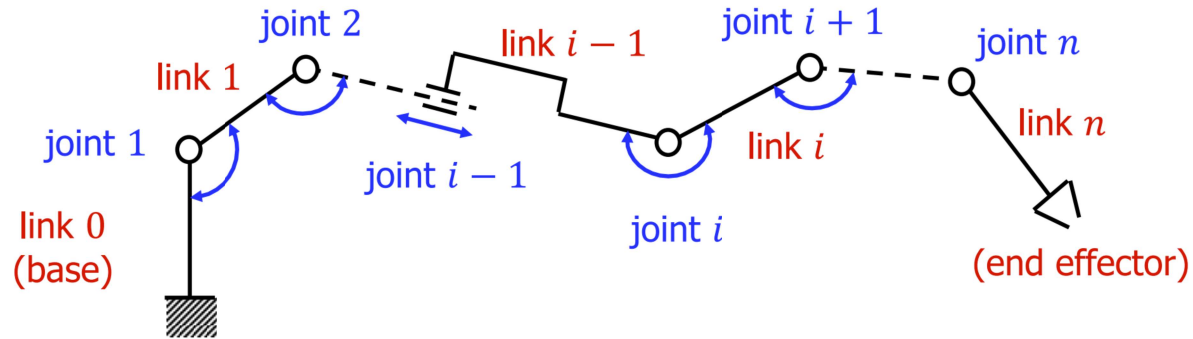
- ▶ What if I fixed only p_x and p_y (with free ϕ)?
- ▶ Our intuition suggests that we have two symmetric configurations of q_1 and q_2 that produce the same (p_x, p_y) .

With some math we can find:

$$q_2 = \pm \arccos \left(\frac{p_x^2 + p_y^2 - l_1^2 - l_2^2}{2l_1 l_2} \right)$$

$$q_1 = \arctan \left(\frac{p_y}{p_x} \right) - \arctan \left(\frac{l_2 \sin(q_2)}{l_1 + l_2 \cos(q_2)} \right)$$

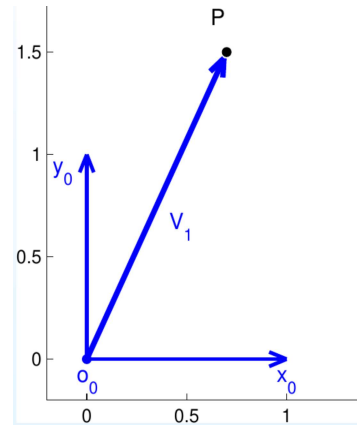
Generally Speaking.....



- ▶ Given a manipulator with joint variables $q = [q_1, q_2, \dots, q_N]^T$. Let p be the pose (position and orientation) of the end effector.
- ▶ The Forward (or Direct) kinematic problem is about finding a function $f(\cdot)$ such that $p = f(q)$.
- ▶ Likewise, the inverse kinematic problem is about finding the function $f^{-1}(\cdot)$ such that $q = f^{-1}(p)$
- ▶ We have seen an adhoc solution for the 2D planar robot.
- ▶ We are now seeking a systematic approach to solve the problem in the general case.

A vector

Let us take a step back.



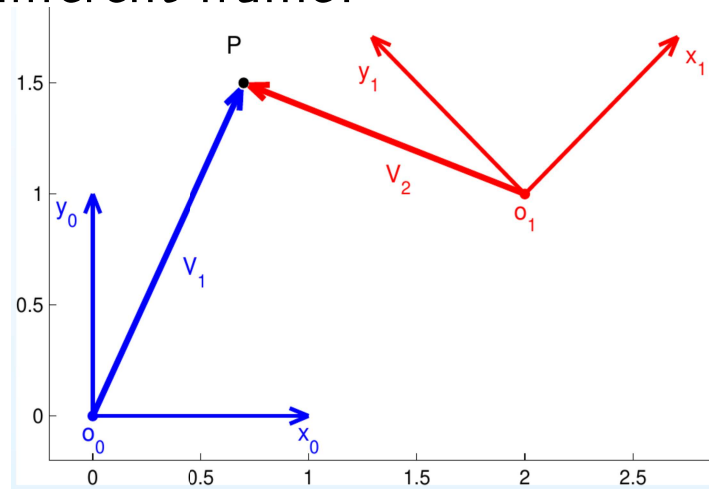
- ▶ Consider a point P . It can be associated with a vector v_1
- ▶ If we fix a frame $O - x_0y_0$ the point P can be associated with the coordinates of the vector in the frame.

$$P^0 = \begin{bmatrix} x_p^0 \\ y_p^0 \end{bmatrix}$$

- ▶ We use the superscript to denote the frame.

Another vector

What if I consider a different frame?



► The same point can be associated with a vector v_2

► So we have

$$P = \begin{bmatrix} x_p^0 \\ y_p^0 \end{bmatrix} == \begin{bmatrix} x_p^1 \\ y_p^1 \end{bmatrix}$$

► We use the superscript to denote the frame.



To summarise

- ▶ A point corresponds to a particular location in space
- ▶ A point can be associated with different vectors and have a different representation (coordinates) in different frames
- ▶ How can we express the coordinates in one frame given the coordinates in another frame?

Example

- ▶ Let us consider two frames with the same origin. The frame $O' - x'y'$ is rotated of α . Consider point P with polar coordinates (ρ, θ) with respect to the $O - xy$ frame

- ▶ It is easy that

$$p_x = \rho \cos(\theta) \quad p_y = \rho \sin(\theta)$$

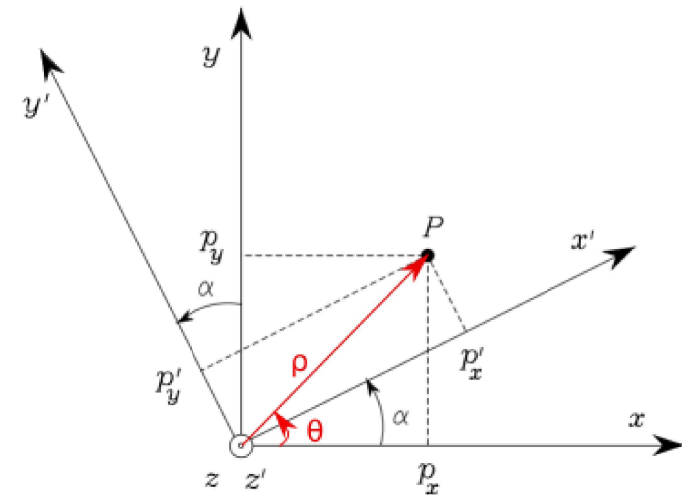
$$p'_x = \rho \cos(\theta - \alpha) \quad p'_y = \rho \sin(\theta - \alpha)$$

- ▶ Applying simple rules on the trigonometric functions, we find:

$$p'_x = p_x \cos(\alpha) + p_y \sin(\alpha)$$

$$p'_y = -p_x \sin(\alpha) + p_y \cos(\alpha)$$

which means $p' = R(\alpha)p$



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$$R(\alpha) = \begin{bmatrix} \cos(\alpha) & \sin(\alpha) \\ -\sin(\alpha) & \cos(\alpha) \end{bmatrix}$$

Example

► Matrix

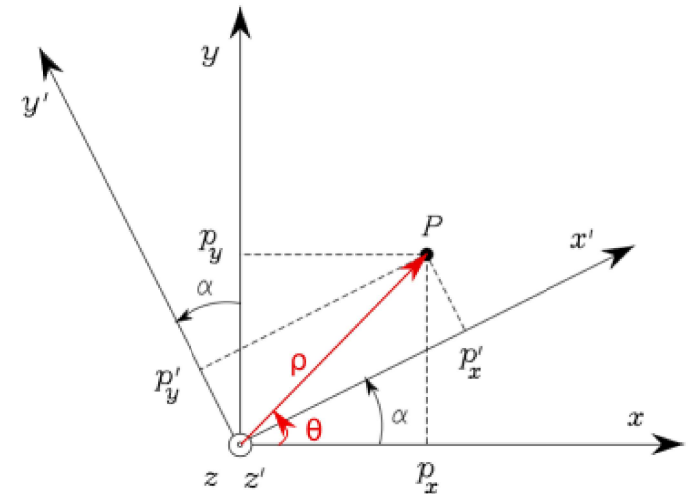
$$R(\alpha) = \begin{bmatrix} \cos(\alpha) & \sin(\alpha) \\ -\sin(\alpha) & \cos(\alpha) \end{bmatrix}$$

is called rotation matrix.

► It is quite easy to see that

$$\det(R(\alpha)) = \cos^2(\alpha) + \sin^2(\alpha) = 1$$

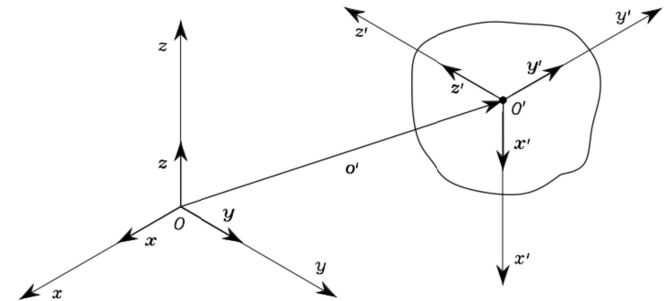
$$R^{-1}(\alpha) = \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) \\ \sin(\alpha) & \cos(\alpha) \end{bmatrix} = R^T(\alpha)$$



► We will soon look at this from a more general 3D perspective.

Pose of a rigid body

- ▶ Robots are modeled as a collection of rigid bodies
- ▶ A preliminary point is defining what we mean by pose.
- ▶ Consider a generic rigid body. Assume that there is a frame attached to the body.
- ▶ By definition of rigid body, the coordinates of the body's points are invariant in the frame attached to the robot.
- ▶ Therefore, the body's configuration in the space is uniquely defined if we define the position of the frame attached to the robot with respect to a fixed frame.



Pose of a rigid body

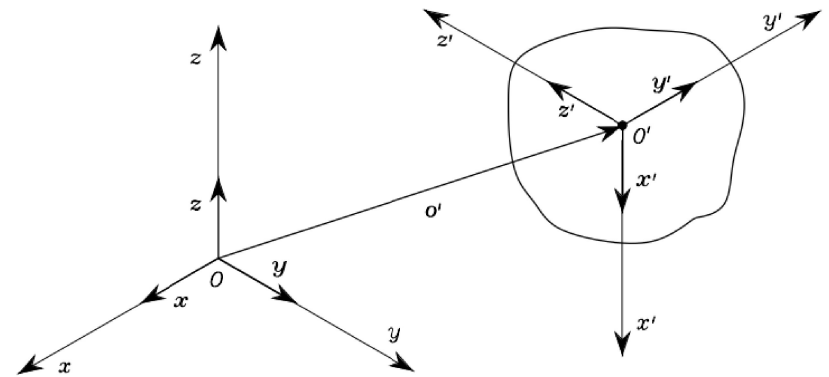
The frame is uniquely identified by the position of its origin (which is a bound vector) and by the coordinates of the three unit vectors of the frame attached to the robot expressed in the world frame.

$$o' = \begin{bmatrix} o'_x \\ o'_y \\ o'_z \end{bmatrix}$$

$$x' = x'_x x + x'_y y + x'_z z$$

$$y' = y'_x x + y'_y y + y'_z z$$

$$z' = z'_x x + z'_y y + z'_z z$$



Rotation Matrix

- ▶ We can equivalently create a matrix with the coordinates of the three unit vectors, which we call *rotation matrix*

Rotation Matrix

$$R = \begin{bmatrix} x'_x & y'_x & z'_x \\ x'_y & y'_y & z'_y \\ x'_z & y'_z & z'_z \end{bmatrix}$$

- ▶ The different elements of R can be found by using the scalar product

$$R = \begin{bmatrix} x'^T x & y'^T x & z'^T x \\ x'^T y & y'^T y & z'^T y \\ x'^T z & y'^T z & z'^T z \end{bmatrix}$$

Rotation Matrix (1)

- Let us compute

$$R^T R = \begin{bmatrix} x'^T x' & y'^T x' & z'^T x' \\ x'^T y' & y'^T y' & z'^T y' \\ x'^T z' & y'^T z' & z'^T z' \end{bmatrix}$$

- Since the frame $O' - x'y'z'$ is made of orthonormal vectors, we have $x'^T x' = y'^T y' = z'^T z' = 1$, while $x'^T y' = x'^T z' = y'^T z' = 0$, which leads to:

$$R^T R = I_3 \rightarrow R^T = R^{-1}.$$

Key property of the Rotation Matrix

The Rotation matrix is an orthogonal matrix with $\det(R) = 1$ for right-handed frames and $\det(R) = -1$ for left-handed frames.

SO(3)

- ▶ The Rotation Matrices have the following properties:
 1. Composing (i.e., multiplying) two Rotation Matrices produces a Rotation Matrix
 2. Rotations respect the associative property:
$$R_1 (R_2 R_3) = (R_1 R_2) R_3$$
 3. Every Rotation has a unique inverse
 4. The degenerate Rotation I_3 is a Rotation
- ▶ Rotations do not commute
- ▶ These property qualify Rotations as an algebraic non-abelian (i.e., non-commutative) group under composition named $SO(3)$

Elementary Rotations

Let us consider a rotation of an angle α about the z axis. It is easy to see that

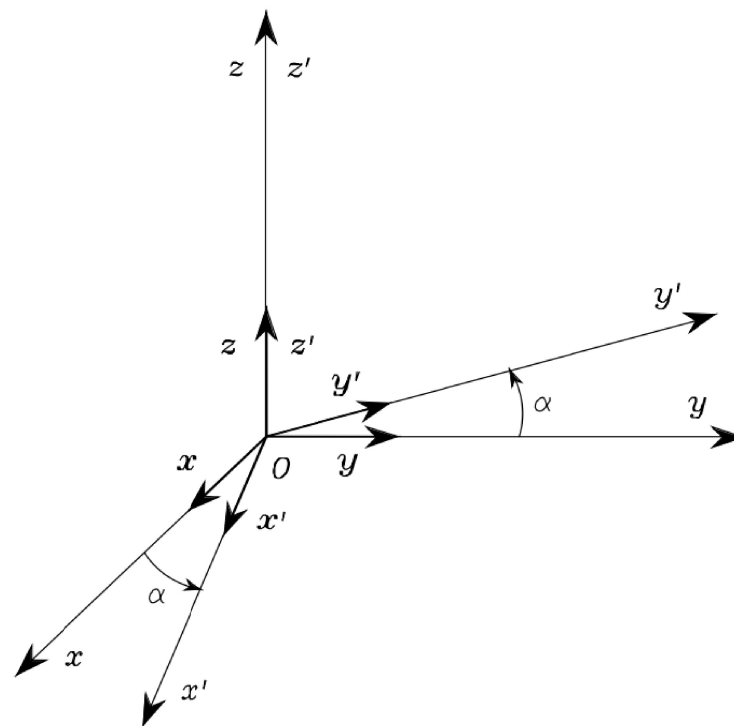
$$x' = \begin{bmatrix} \cos(\alpha) \\ \sin(\alpha) \\ 0 \end{bmatrix}$$

$$y' = \begin{bmatrix} -\sin(\alpha) \\ \cos(\alpha) \\ 0 \end{bmatrix}$$

$$z' = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

with a resulting matrix :

$$R_z(\alpha) = \begin{bmatrix} \cos(\alpha) & -\sin(\alpha) & 0 \\ \sin(\alpha) & \cos(\alpha) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$





Elementary Rotations (1)

- Likewise, we can find the rotation matrix of an angle β about y and of an angle γ about x

$$R_y(\beta) = \begin{bmatrix} \cos(\beta) & 0 & \sin(\beta) \\ 0 & 1 & 0 \\ -\sin(\beta) & 0 & \cos(\beta) \end{bmatrix}$$

$$R_x(\gamma) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\gamma) & -\sin(\gamma) \\ 0 & \sin(\gamma) & \cos(\gamma) \end{bmatrix}$$

- We can easily verify

$$R_k(-\theta) = R_k^T(\theta), \quad k = x, y, z.$$



Geometric Interpretation

- ▶ In view of what we have just seen, we can provide a geometric interpretation of rotation matrices.

Geometric interpretation of rotation matrices

A rotation matrix R represents a rotation about an axis in space needed to align the axes of the reference frame with the corresponding axes of the body frame.

Representation of a Vector

- ▶ Let us consider the case in which the two origins O and O' coincide.
- ▶ A generic point can be represented by

$$\mathbf{p} = \begin{bmatrix} p_x \\ p_y \\ p_z \end{bmatrix} \quad \text{in the } O - xyz \text{ frame}$$

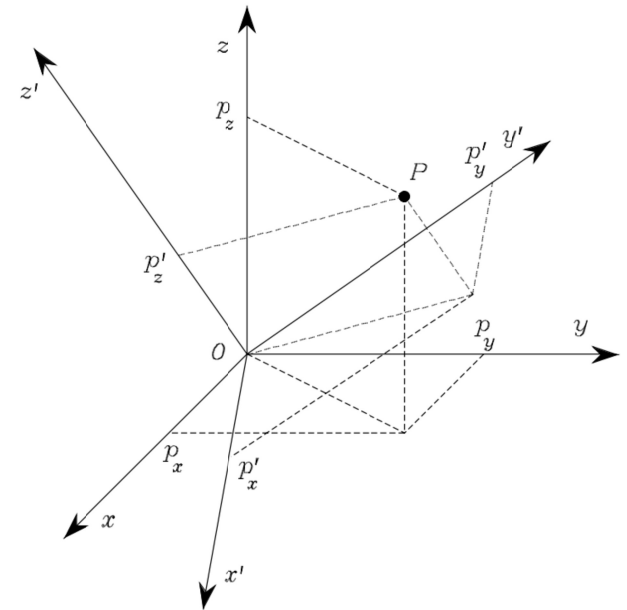
$$\mathbf{p}' = \begin{bmatrix} p'_x \\ p'_y \\ p'_z \end{bmatrix} \quad \text{in the } O - x'y'z' \text{ frame}$$

- ▶ Since both $\mathbf{p}$ and $\mathbf{p}'$ refer to the same point, we can write:

$$\mathbf{p} = p'_x \mathbf{x}' + p'_y \mathbf{y}' + p'_z \mathbf{z}' = [\mathbf{x}' \ \mathbf{y}' \ \mathbf{z}'] \mathbf{p}' = R \mathbf{p}'$$

and exploiting the properties of R :

$$\mathbf{p} = R^T \mathbf{p}'$$



Example

- ▶ Let us consider two frames with the same origin. The frame $O' - x'y'z'$ is rotated of α about the z axis. Consider point P with polar coordinates (ρ, θ) with respect to the $O - xyz$ frame

- ▶ It is easy that

$$p_x = \rho \cos(\theta) \quad p_y = \rho \sin(\theta)$$

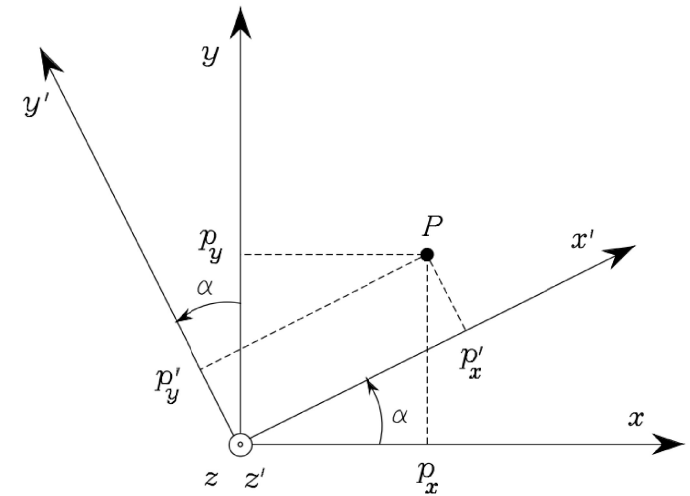
$$p'_x = \rho \cos(\theta - \alpha) \quad p'_y = \rho \sin(\theta - \alpha)$$

- ▶ Applying simple trigonometric functions, we find:

$$p'_x = p_x \cos(\alpha) + p_y \sin(\alpha)$$

$$p'_y = -p_x \sin(\alpha) + p_y \cos(\alpha)$$

$$p'_z = p_z$$



Rotation of a Vector

- ▶ Another way of seeing a rotation matrix is as a matrix operator enabling the rotation of a vector about an arbitrary axis in the space
- ▶ Consider a rotation matrix R . and let p' be a vector expressed in the $O - xyz$ and the vector $p = Rp'$. We can easily see:

$$\begin{aligned}\|p\|^2 &= p^T p \\ &= p'^T R^T R p \\ &= p'^T p\end{aligned}$$

- ▶ The rotation operator preserves the norm

Example

Consider the vector obtained rotating a vector p' of an angle α about the z -axis. Suppose that the vector p' has polar coordinates (ρ, θ) . Vector p will have polar coordinates $(\rho, \alpha + \theta)$. We have:

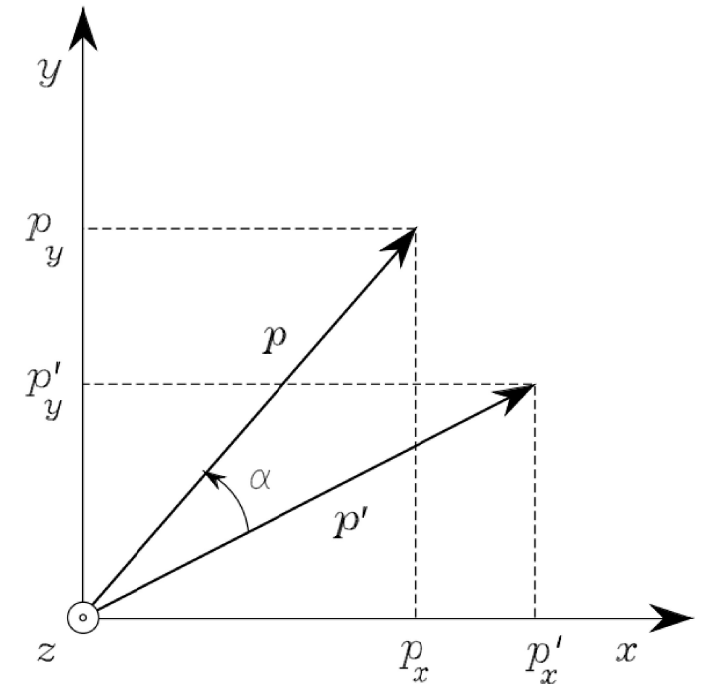
$$\begin{aligned} p'_x &= \rho \cos(\theta) & p'_y &= \rho \sin(\theta) \\ p_x &= \rho \cos(\theta + \alpha) & p_y &= \rho \sin(\theta + \alpha) & p_z &= p'_z \end{aligned}$$

With little effort, we find:

$$\begin{aligned} p_x &= p'_x \cos(\alpha) - p'_y \sin(\alpha) \\ p_y &= p'_x \sin(\alpha) + p'_y \cos(\alpha) \\ p_z &= p'_z \end{aligned}$$

that means

$$p = R_z(\alpha)p'.$$





Summary

- ▶ We have seen three possible meanings for rotation matrices

Geometric Meanings of a Rotation Matrix

1. It describes the mutual orientation between two coordinated frames. Its column vectors are the direction cosines of the axes of the rotated frame with respect to the original one.
2. It represents the coordinate transformation between two rotated frames
3. It is the operator that enables rotations within the same coordinate frame